From this consideration, two central things become clear. First and most central, it is possible to obtain solutions that are oscillatory in one far field with both a complex wavenumber and a complex temporal frequency. Second, for these modes, the question of finding eigenvalues to appropriate BCs (analogous to the BC set in chapter 5), the 4D parameter space for a given Mach number reduces by equations (7.2) or (7.5) to a 3D parameter space, finding values of $\omega_r, \alpha_r, \omega_i$ that allow power series matching as described in chapter 3. It should be emphasized that the modes considered in this section are of particular interest because they combine far-field radiation with a possible instability. However, as discussed above, assessing them as stable or unstable is not trivial. Therefore, Briggs' stability criterion was established,the derivation of which is reviewed in the next section.

7.2. Reviewing Briggs method and the concept of convective and absolute instabilities

In this section, we give an overview onf the derivation of criteria for absolute and convective instabilities proposed by Briggs (1964). We mainly follow Briggs' derivation here, but for readers who are not familiar with complex analysis in, some basic concepts are additionally reviewed, because they are only briefly mentioned in Briggs' work, but not explained in detail. The basic idea is to consider the response of the flow to an external disturbance. For this, normal mode solutions have to be super-positioned, which is motivated in the following.

Remark why a single mode consideration is not physical if spatial growth is considered

We consider the temporal evolution of a response, i.e. we assume that disturbances grow from their initial state until the limit of linear stability theory is reached when the response becomes that large that non-linear effects must be taken into account. However, only finite flow quantities are physically plausible. This is the reason why it makes no sense on an infinite domain $x \in (-\infty, \infty)$ to consider only a single mode of type

$$q'(x, y, t) = \hat{q}(y) e^{i(\alpha x - \omega t)}, \tag{7.8}$$

with $\alpha \in \mathbb{C}$ and $\omega \in \mathbb{R}$. It is obvious that, due to the unbounded domain, the consideration of this single mode violates the assumptions behind linear stability theory, since such a solution already has an infinite value on the domain at initial time $t = 0$. Therefore, we have to consider the superposition of all modes to obtain a meaningful response and each solution of the dispersion relation contributes to that response.

Distinguishing absolute and convective instabilities

We classify each solution of the dispersion relation (and accordingly each eigenvalue) based on the question in which way it contributes to the long-term response of the flow to an external source. We distinguish between convective and absolute instabilities. This distinction was first introduced by Sturrock (1958). We consider an instability as being absolute if, in an infinite system, a pulse disturbance which is spatially finite grows in time without limit at every point in space. However, if a disturbance grows in time but propagates spatially while it evolves, it is possible that its amplitude decreases at any fixed point in space and increases only along some characteristic $x(t)$. It is then called a convective instability. In other words, once an absolutely unstable disturbance has reached a certain point, the disturbance will at this point keep growing in time. In contrast to that, a convective instability will at a fixed point lead to growth first but then to decay. This is visualized in figure 7.1.

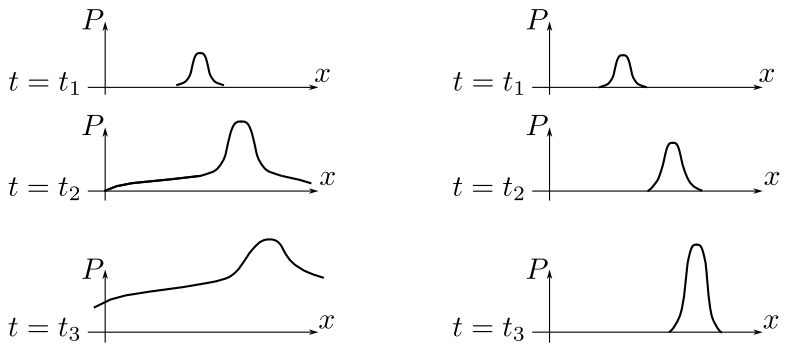

Figure 7.1.: Visualization of the difference between absolute (left) and convective instabilities (right). The evaluation of an unstable perturbation P in space is shown for three different time steps $t_1 < t_2 < t_3$. The illustration is similar to that given by Briggs (1964).

7.2.1. Green's function formalism leading to the response to a source

Eigenvalues emerge as a solution of the so-called dispersion relation, in a sense including both the solution of the differential equation and the BC. For the power series matching principle established in this work, the dispersion relation can always be written in the form of (3.39). Formally, we express the dispersion relation as

$$\Delta(\omega, \alpha) = 0. \tag{7.9}$$

In general, all solutions of (7.9) will contribute to the homogeneous response to a localized source. To formally express the response of the flow to such a source, we extend the 2D linearized Euler operator, given by (2.14) and formally written as $\mathcal{L}(P) = 0$, where P summarizes the perturbations. (2.14) is augmented with a source s to

$$\mathcal{L}(P) = s_x(x)s_t(t)s_y(y), \tag{7.10}$$

assuming that the source of variable-separation form. Then, P denotes the response of the flow to this source, i.e. a solution of the inhomogeneous linearized Euler equations. Given a Green's function G, which is the response to an impulse disturbance (see A.6.4), we express P as

$$P(x,t,y) = \int_{-\infty}^{\infty} G(x-x', t-t', y, y')\, s_x(x')s_t(t')s_y(y')\, \mathrm{d}x'\mathrm{d}t'\mathrm{d}y' \tag{7.11}$$

Note that homogeneity (or in other words invariance under translations) in x and t allowed us to write G depending on $x - x'$ and $t - t'$. Further, for reasons of causality, we assume that the source vanishes for $t < 0$. We will refer to this point later. In a next step, we carry out Fourier and Laplace transformations of (7.11). Fourier transform of a function $f(x)$ and its inverse are defined in accordance with the normal mode approach (7.8) by

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\alpha)e^{i\alpha x}\mathrm{d}\alpha, \qquad f(\alpha) = \int_{-\infty}^{\infty} f(x)e^{-i\alpha x}\mathrm{d}x. \tag{7.12}$$

We further define the (rotated) Laplace transform of a function $g(t)$ by

$$g(t) = \frac{1}{2\pi}\int_{-\infty+i\sigma}^{\infty+i\sigma} g(\omega)e^{-i\omega t}\mathrm{d}\omega, \qquad g(\omega) = \int_{0}^{\infty} g(t)e^{i\omega t}\mathrm{d}t. \tag{7.13}$$

Note that for the considerations in this chapter we refrain from introducing different variables for the Fourier/Laplace transformed and the original variable. Instead, the arguments of the functions make

clear to which we refer in the current context, as done in (7.12) and (7.13). The integration contour in the Laplace transform was shifted away from the real axis to the upper complex half plane, ensuring causality. Causality means that we have to demand that the response vanishes for $t < 0$. We will see later why this leads to the shift in the upper complex half plane. Applying Fourier and Laplace transforms to (7.11) yields

$$P(\alpha, \omega, y) = \int_{-\infty}^{\infty} G(\alpha, \omega, y, y') s_y(y') \mathrm{d}y' s_x(\alpha) s_t(\omega) \,, \tag{7.14}$$

where $G(\alpha, \omega, y, y')$ is the Fourier and Laplace transform of the Green function. The result stems from the fact that (7.11) represents a convolution of G with the source functions s_x and s_t - as explained in A.6.1, applying Fourier or Laplace transforms to a convolution integral leads in Fourier/Laplace space to the product of the transformed functions (Meyberg & Vachenauer, 2006). Introducing in accordance with the work of Briggs (1964) the weighted Green's function

$$\hat{G}(\alpha, \omega, y) = \int_{-\infty}^{\infty} G(\alpha, \omega, y, y') s_y(y') \mathrm{d}y', \tag{7.15}$$

and applying inverse Fourier and Laplace transform to (7.14), defined by (7.12) and (7.13), we end up with the spatio-temporal response to a source, given by

$$P(x, t, y) = \frac{1}{4\pi^2} \int_{-\infty+i\sigma}^{\infty+i\sigma} \int_{-\infty}^{\infty} \hat{G}(\alpha, \omega, y) s_x(\alpha) s_t(\omega) e^{i(\alpha x - \omega t)} \mathrm{d}\alpha \mathrm{d}\omega, \tag{7.16}$$

This equation is central for further analysis. With the transformations, we note that the source response is expressed through the Green's function of the operator $\mathcal{L}$. We will see how knowledge of the homogeneous solution transfers to Green's function and consequently to the source response. The goal is to explore the long term behavior of (7.16) for $t \to \infty$ at all points x in order to assess whether this response to a source shows an absolute or convective instability as visualized in figure 7.1.

7.2.2. Analysis of the source response

We will now analyze (7.16). Crucial for this is choice of the respective integration paths. For this integration, the singularities (or poles, which will be used as equivalent wording in this chapter, excluding essential singularities) of the integrand of (7.16) are decisive, since they determine whether and in which way the paths of integration can be deformed. The reason for these singularities in the Green's function in the integrand of (7.16) will be explained later. For now, we just assume that these singularities exist. The path deformation around the singularities will allow us to make statements in the sense of absolute or convective instabilities. We consider integration in frequency and wavenumber domain separately by introducing

$$\hat{P}(x, \omega, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{G}(\alpha, \omega, y) s_x(\alpha) e^{i\alpha x} \mathrm{d}\alpha \tag{7.17}$$

and

$$P(x, t, y) = \frac{1}{2\pi} \int_{-\infty+i\sigma}^{\infty+i\sigma} \hat{P}(x, \omega, y) s_t(\omega) e^{-i\omega t} \mathrm{d}\omega. \tag{7.18}$$

To obtain the source response, these equations can be understood as follows: For each frequency on the inverse Laplace contour (7.18), we carry out the integration along the inverse Fourier contour on the real α-axis (7.17).

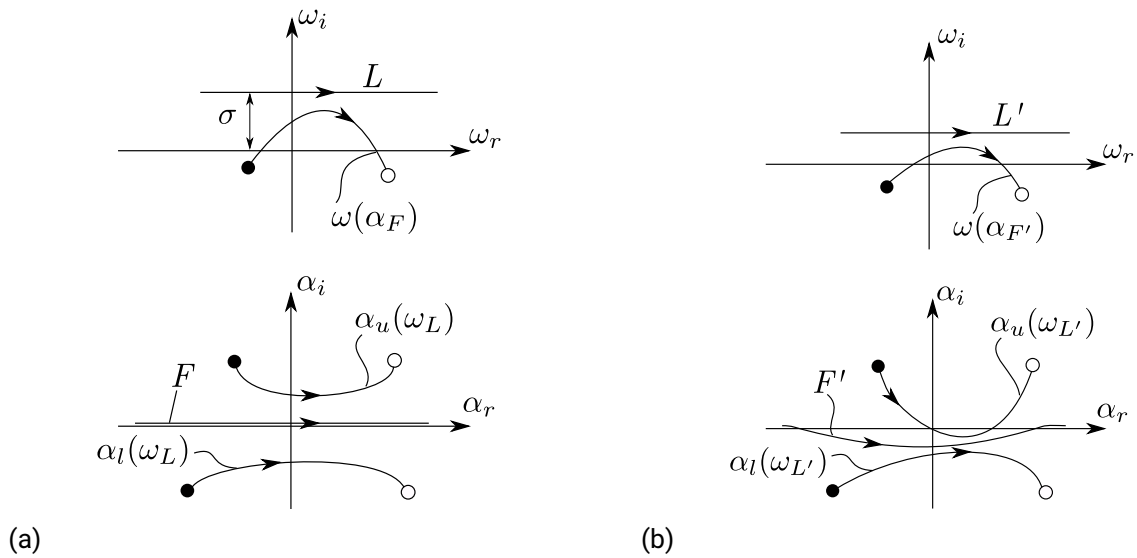

Figure 7.2.: Deformation of the paths of integration of the inverse Laplace and Fourier integrals. (a) shows the original paths and (b) the deformed ones.

Path deformation

The aforementioned deformation process can be done without changing the resulting source response if the singularities of the integrand above and below the paths of integration remain the same, which follows from the residue theorem. We refer to the residue theorem in more detail later and it is discussed in appendix A.6.2. Figure 7.2 shows a deformation of both the inverse Fourier (denoted as F) and Laplace contour (L) (figure 7.2a) to deformed curves F' and L' (figure 7.2b).

We exemplary show the trajectories of two poles in the upper and lower complex α half plane (denoted by α_u and α_l) and the trajectory of a pole in the complex ω plane. The original inverse Laplace contour L is chosen in a way that it both lies above the most unstable eigenvalue for $\alpha \in \mathbb{R}$ and that the trajectories of the poles in the complex α plane do not cross the real axis in order to keep the integrand analytic along the inverse Fourier transform (7.17).

Let us further discuss how the trajectories of the poles are formed: When integrating (7.16) along the inverse Fourier curve F, the trajectory of the pole in the ω - plane emerges as $\omega(\alpha_F)$. Since the ordering of integration is arbitrary by Fubini's theorem, in the same way, the trajectories of the poles in the α - plane emerge from integrating along the inverse Laplace contour L as $\alpha(\omega_L)$. Consequently, the trajectory of both the poles in the ω and α plane changes under a deformation of the paths of integration, which is shown in figure 7.2b. We will in the following discuss the influence of the different trajectories of the poles on the representation of the source response (7.16). Therefore, we discuss the integration in α and ω space separately.

Inverse Laplace contour

Let us first discuss why the path of integration in the inverse Laplace transform (7.18) is chosen to lie at $\omega_i = \sigma$ and why σ has to be larger than the most unstable eigenvalue occurring for $\alpha \in \mathbb{R}$.

The first requirement on $P(x, t, y)$ is causality. Since we assume that the source vanishes for $t < 0$, we also have to demand for the response that (7.18)$= 0$ for $t < 0$. To ensure this, we consider the singularities of $P(x, \omega, y)$ in the complex ω-plane arising along the inverse Fourier contour, i.e. for

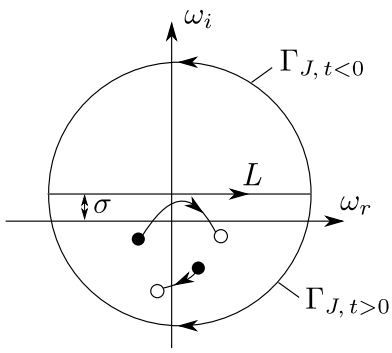

Figure 7.3.: Closing the inverse Laplace contour L for both $t < 0$ and $t > 0$ as an extension of figure 7.2a. Poles of the integrand arising for $\alpha \in \mathbb{R}$ are marked as dots and their trajectory is shown for a variation of α along the real axis.

real α in (7.17). The singularities marked as dots in figure 7.3 and exemplary trajectories of the singularities, emerging when α is varied along F, are shown. We chose the value of σ in a way that the path of integration lies over all singularities. This allows us to close the contour in the upper complex ω half-plane for the case $t < 0$. Since no singularities are enclosed in this contour, we conclude from the residue theorem (see A.6.2) for the closed contour that

$$\frac{1}{2\pi}\int_{-\infty+i\sigma}^{\infty+i\sigma} \hat{P}(x,\omega,y)s_t(\omega)e^{-i\omega t}\mathrm{d}\omega + \frac{1}{2\pi}\int_{\Gamma_J} \hat{P}(x,\omega,y)s_t(\omega)e^{-i\omega t}\mathrm{d}\omega = 0, \tag{7.19}$$

holds. For $t < 0$, we can conclude from Jordan's lemma (see appendix A.6.3) that the second term in (7.19) vanishes and therefore also the first term vanishes which represents the response (7.18). Consequently, we retain causality and the response vanishes for $t < 0$. Note that for the previous considerations, we assumed analyticity of the Laplace and Fourier transformed sources $s_t(\omega)$ and $s_x(\alpha)$ in the closed contour, i.e. for $\omega_i > \sigma$. Since we will be considering impulse sources in this chapter, we were allowed to do so.

As mentioned above, singularities of Green's function arising for real wavenumbers determine the shift of the path of integration in (7.18). For an ODE of second order, it can be shown that these singularities are identical with the eigenvalues of the problem, i.e. the solutions of the dispersion relation (7.9) (Teschl, 2012). This is summarized and explained in more detail in A.6.4. We can thus conclude that σ in (7.18) has in any case to be chosen larger than the largest ω_i that occurs for $\alpha \in \mathbb{R}$. Secondly, σ has to be large enough that integration along F laying on the real axis can be carried out. In other words, we have to ensure that $\alpha(\omega_F)$ in any case has an imaginary part.

In contrast to that, for $t > 0$ the inverse Laplace contour has to be closed in the lower complex ω half plane in order to apply Jordan's lemma. Therefore, all poles included in that contour closing contribute to the response for $t > 0$. Consequently, the pole with largest temporal growth rate ω_i contributing is the one with largest ω_i for <u>real</u> α.

A further remark should be made on the case where only eigenvalues with $\omega_i < 0$ occur for $\alpha \in \mathbb{R}$, which is typically referred to as temporally stable. Then, the inverse Laplace contour L can be put below the real ω axis (meaning that $\sigma < 0$) and closed in the lower half ω-plane. After applying Jordan's

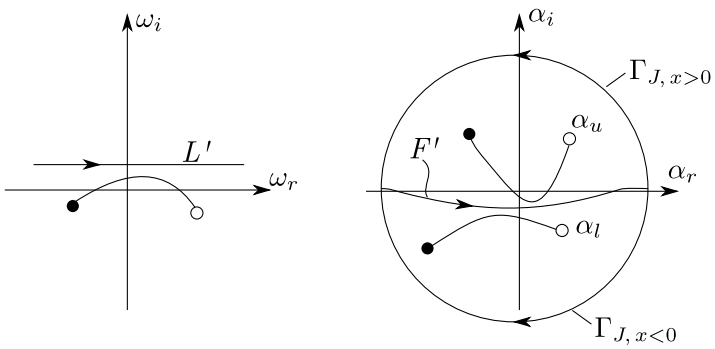

Figure 7.4.: Inverse Fourier contour closing (right) for $x > 0$ and $x < 0$ for the Laplace contour (left) lowered as far as possible, including the contours of the poles, which are marked as dots.

lemma and the residue theorem (see A.6.2), we find for $t > 0$ the source response

$$P(x,t,y) = \frac{1}{2\pi}\int_{-\infty+i\sigma}^{\infty+i\sigma} \hat{P}(x,\omega,y)s_t(\omega)e^{-i\omega t}\mathrm{d}\omega$$
$$= 2\pi i \sum_n \mathrm{ind}_\Gamma(\omega_{0,n})\mathrm{Res}_{\omega_{0,n}}\left(\frac{\hat{F}(x,\omega,y)s_t(\omega)e^{-i\omega t}}{2\pi}\right), \tag{7.20}$$

where the summation is over all n poles $\omega_{0n,}$ in the lower half plane. Considering the asymptotic limit $t \to \infty$, it becomes clear that this source response is decaying, clearly showing that a flow having only eigenvalues in the lower complex ω half plane has to be considered as stable.

Inverse Fourier contour

As pointed out previously, the singularities in the ω plane are decisive for the response. However, the question might arise as to how the singularities occur. The key point in understanding this lies in the deformation of the paths of integration as shown in figure 7.2. Deforming the paths L and F to L' and F' as sketched in figure 7.4 is possible until the L contour can not be lowered any further without violating causality, i.e. until the L contour lies just right above the largest ω_i. We may thus deform the paths of integration to that stage and see which insight we get. To understand the spatial behavior of the response, included in the inverse Fourier integral (7.17), we analyze this integral expression for the deformed contours. For the situation sketched in figure 7.2, let us consider the integration in the α plane, namely the inverse Fourier integral given by

$$\hat{P}(x,\omega,y) = \frac{1}{2\pi}\int_{F'} \hat{G}(\alpha,\omega,y)s_x(\alpha)e^{i\alpha x}\mathrm{d}\alpha \tag{7.21}$$

The former expression contains the x-dependence of the response and the Fourier transform of the source s_x, which we assume without loss of generality to be at $x = 0$. In order to evaluate (7.21), we close the integration contour and apply both Jordan's lemma (see A.6.3) and the residue theorem (A.6.2). If the contour is closed in a way that the semicircle vanishes due to Jordan's lemma, then, according to the residue theorem, the value of (7.21) follows only from the poles included in the closed path. This is carried out in more detail in the following.

If we consider $x > 0$, the contour of integration is closed as shown in the upper complex α half plane by the semicircle $\Gamma_{J,x>0}$. For $x < 0$, we close the contour in the lower half plane by $\Gamma_{J,x<0}$. Then, we

extend (7.21) and obtain for the integral along the closed contour

$$\frac{1}{2\pi}\int_{F'}\hat{G}(\alpha,\omega,y)s_x(\alpha)e^{i\alpha x}\mathrm{d}\alpha+\frac{1}{2\pi}\int_{\Gamma_J}\hat{G}(\alpha,\omega,y)s_x(\alpha)e^{i\alpha x}\mathrm{d}\alpha$$
$$=2\pi i\sum_n \mathrm{ind}_\Gamma(\alpha_{0,n})\mathrm{Res}_{\alpha_{0,n}}\left(\frac{\hat{G}(\alpha,\omega,y)s_x(\alpha)e^{i\alpha x}}{2\pi}\right). \tag{7.22}$$

Again, we applied the residue theorem (see A.6.2) and have to sum over all n poles $\alpha_{0,n}$ that are included in the respective closed contour as shown in figure 7.4. There, they are exemplary given by α_u for the closure in case $x>0$ and α_l in case $x<0$. From Jordan's lemma, we further conclude that for $x\lesseqgtr 0$ the integral along the semicircle $\Gamma_{J,x\lesseqgtr 0}$ vanishes and thus (7.22) is equivalent to (7.21). Consequently, poles included in the upper closed contour $F'+\Gamma_{J,x>0}$ contribute to the answer in the region $x>0$. Poles included in $F'+\Gamma_{J,x<0}$ contribute to the answer for $x<0$. As the poles of Green's function (or eigenvalues of the dispersion relation) are roots of $\hat{G}^{-1}$, we can rewrite the RHS of (7.22), assuming analyticity of the source s_x, according to (Meyberg & Vachenauer, 2006)

$$\mathrm{Res}_{\alpha_{0,n}}\left(\frac{s_x(\alpha)e^{i\alpha x}}{\left(\hat{G}(\alpha,\omega,y)\right)^{-1}}\right)=\frac{s_x(\alpha_{0,n})e^{i\alpha_{0,n}x}}{\left(\frac{\partial}{\partial\alpha}\left(\hat{G}(\alpha,\omega,y)\right)^{-1}\right)\Big|_{\alpha_{0,n}}}, \tag{7.23}$$

and therefore obtain for (7.21), which is equivalent to (7.17),

$$\hat{P}(x,\omega,y)=\frac{1}{2\pi}\int_{F'}\hat{G}(\alpha,\omega,y)s_x(\alpha)e^{i\alpha x}\mathrm{d}\alpha=\sum_n\frac{\pm i\,s_x(\alpha_{0,n})e^{i\alpha_{0,n}x}}{\left(\frac{\partial}{\partial\alpha}\left(\hat{G}(\alpha,\omega,y)\right)^{-1}\right)\Big|_{\alpha_{0,n}}}, \tag{7.24}$$

where the $\pm$ sign follows from the orientation of $\Gamma+F'$. The $+$ sign holds when considering the answer for $x>0$ and a counterclockwise oriented contour, i.e. for poles in the upper contour closure. The $-$ sign follows for $x<0$, i.e. poles in the lower clockwise oriented contour closure.

7.2.3. Obtaining criteria for convective and absolute instabilities

We may now analyze the source response with the aim of revealing its growth in both space and time. This is done for the quite general pulse source, given by delta distributions in the physical space. Green's function is exactly defined as the response to this kind of source (see A.6.4). Thus, assuming the source to be of the type

$$s_x(x)=\delta(x),\qquad s_x(\alpha)=1, \tag{7.25}$$
$$s_t(t)=\delta(t),\qquad s_t(\omega)=1, \tag{7.26}$$

we obtain from (7.16), with the deformed curves, the response as the inverse Fourier and Laplace transform of a (weighted) Green's function, given by

$$P(x,t,y)=\frac{1}{4\pi^2}\int_{L'}\int_{F'}\hat{G}(\alpha,\omega,y)e^{i(\alpha x-\omega t)}\mathrm{d}\alpha\mathrm{d}\omega. \tag{7.27}$$

As mentioned above, the contour L' can for reasons of causality/analyticity be lowered until it is just above the pole with largest ω_i arising for the corresponding deformed contour F'. During this process of lowering the contour, different loci of the poles might arise, leading to either absolute or convective instabilities.

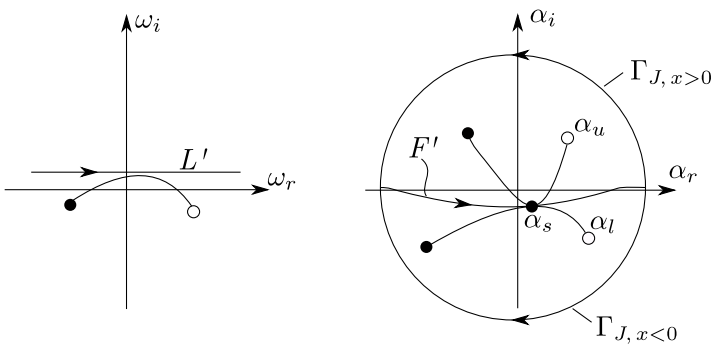

Figure 7.5.: Deformed inverse Laplace and Fourier contours L' and F' for poles in the α merging into a pole of order 2.

Absolute instabilities

Following Briggs' argument, the inverse Fourier path deformation keeping invariant the number of poles of Green's function above and below F' in order to preserve (7.21) fails in the case that two poles merge into each other at $\omega = \omega_s$ and $\alpha = \alpha_s$ as exemplary sketched in figure 7.5. If these poles merge from two different sides of F', the contour of integration can not be drawn around these poles. Merging of the poles means that they are at the same time double α-roots of the inverse weighted Green's function $\hat{G}^{-1}$ and it therefore holds both

$$\hat{G}^{-1}(\alpha_s, \omega_s, y) = 0, \qquad \text{and} \qquad \left.\frac{\partial \hat{G}^{-1}}{\partial \alpha}\right|_{\alpha_s,\omega_s} = 0. \tag{7.28}$$

From these conditions, it becomes clear why this case is referred to as absolute instability. Considering the representation of the Laplace transform of the source response given by (7.24), this becomes for the pulse source

$$\hat{P}(x, \omega, y) = \sum_n \frac{i\, e^{i\alpha_{0,n} x}}{\left.\left(\frac{\partial}{\partial \alpha}\left(\hat{G}(\alpha, \omega, y)\right)^{-1}\right)\right|_{\alpha_{0,n}}}. \tag{7.29}$$

Note that the $\pm$ sign is omitted from now on since it is not decisive for the stability classification. We observe two central aspects: First, the response has exponentially growing behavior in space: Since the pole included in the upper contour has $\alpha_i < 0$, i.e. is below the real α axis, it leads to exponential growth in x for $x > 0$. Second and most central, with (7.28) we find the response to have a singularity as the poles reach the limit $\alpha_{0,n}, \omega \to \alpha_s, \omega_s$. In other words, the solution "blows up" once the poles /eigenvalues merge into each other, which we call an absolute instability.

At this point, two things should be mentioned. First, such an absolute instability is only present if the poles merge from above and below of F', meaning that one must have crossed the real α axis during the path deformation. Second, as the poles are nothing but roots of the dispersion relation (7.9), we may also rewrite (7.28) in terms of the dispersion relation. A double root of the dispersion relation at ω_s, α_s is nothing but a saddle point given by

$$\Delta(\omega_s, \alpha_s) = 0, \qquad \text{and} \qquad \left.\frac{\partial \Delta(\omega, \alpha)}{\partial \alpha}\right|_{\alpha_s,\omega_s} = 0. \tag{7.30}$$

Differentiation of the dispersion relation $\Delta(\omega, \alpha) = 0$ w.r.t. α then leads, using (7.30), at the saddle point to

$$\left(\frac{\partial\omega}{\partial\alpha}\frac{\partial\Delta}{\partial\omega} + \frac{\partial\Delta}{\partial\alpha}\right)_{\alpha_s,\omega_s} = \left.\frac{\partial\omega}{\partial\alpha}\right|_{\alpha_s,\omega_s} = 0, \tag{7.31}$$

which is a necessary criterion for the occurrence of an absolute instability at α_s, ω_s. However, the path deformation has to be carried out for each concrete case to obtain a sufficient statement. If $\partial\omega/\partial\alpha$ is referred to as group velocity, a zero group velocity is the necessary condition for an absolute instability. In order to further assess the behavior of the response in space included in (7.29) for an absolute instability, we may use (7.28) to expand $\hat{G}^{-1}$ (or the dispersion relation Δ) as Taylor series, which is in the leading orders given by

$$\hat{G}^{-1}(\alpha_s, \omega_s, y) \approx \left.\frac{\partial\hat{G}^{-1}}{\partial\omega}\right|_{\alpha_s,\omega_s} (\omega - \omega_s) + \frac{1}{2}\left(\frac{\partial^2\hat{G}^{-1}}{\partial\alpha^2}\right)_{\alpha_s,\omega_s} (\alpha - \alpha_s)^2. \tag{7.32}$$

According to Briggs, we use this in (7.29) and subsequently consider the limit $\omega \to \omega_s$ in order to obtain the response. This leads to

$$\hat{P}(x, \omega, y) \approx \frac{e^{i\alpha_s x}}{\left.\left(2\left(\frac{\partial^2\hat{G}^{-1}}{\partial\alpha^2}\right)\left(\frac{\partial\hat{G}^{-1}}{\partial\omega}\right)\right)^{1/2}\right|_{\alpha_s,\omega_s}} \cdot (\omega - \omega_s)^{-1/2}, \tag{7.33}$$

where details on the calculation at this point can be found in the work of Derfler (1962). This result shows both spatial growth away from the disturbance as well as a branch-pole singularity in $\omega \to \omega_s$.

In order to consider the response in both space and time domain, we use (7.33) in (7.27), leading to

$$P(x, t, y) \approx \frac{e^{i\alpha_s x}}{\left.\left(2\left(\frac{\partial^2\hat{G}^{-1}}{\partial\alpha^2}\right)\left(\frac{\partial\hat{G}^{-1}}{\partial\omega}\right)\right)^{1/2}\right|_{\alpha_s,\omega_s}} \cdot \frac{1}{2\pi} \cdot \int_{L'} (\omega - \omega_s)^{-\frac{1}{2}} e^{-i\omega t} \mathrm{d}\omega, \tag{7.34}$$

clearly showing due to $\alpha_s \in \mathbb{C}$ that the response has exponential behavior in space. If the merging of the poles happens in the lower complex α half plane, the response grows exponentially away from the source for $x > 0$ and vice versa. Due to the branch pole at $\omega = \omega_s$ and the respective branch cut, stemming from the square root function, this integral requires an integration by the method of keyhole path deformation to circumvent the branch cut (Meyberg & Vachenauer, 2006). For the sake of brevity, this will not be shown here and the reader is referred to the literature, since this last step in particular requires the consideration of the concrete location of the poles and possibly further branch cuts, and therefore this step is strongly problem dependent. Assuming that only this single branch pole with its corresponding branch cut is present, we obtain for the long-term behavior of the response

$$P(x, t, y) \approx e^{i(\alpha_s x - \omega_s t)} t^{-\frac{1}{2}}, \tag{7.35}$$

which is exponentially growing in time for all x, thus representing an absolute instability. The shape in space is exponential, with growth to the right-hand or left-hand side of the source, depending on the sign of $\Im(\alpha_s)$ as explained above.

Convective instabilities

In the former section, we have found that absolute instability arises due to the merging of poles α_s with $\Im(\omega_s) > 0$. The corresponding trajectories of poles were obtained by contour deformation, especially by

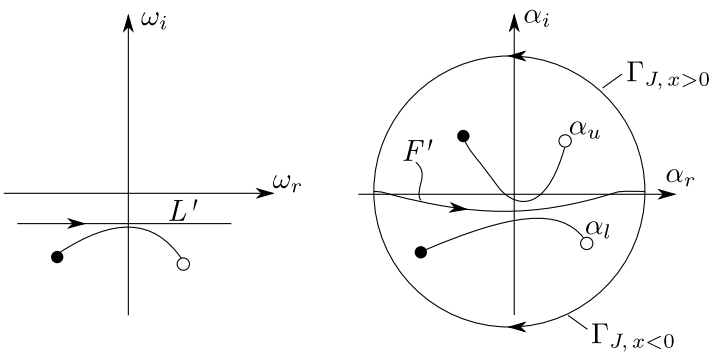

Figure 7.6.: Deformed inverse Fourier and Laplace contour in case of a convective instability.

lowering the inverse Laplace contour as shown in figure 7.5. There, the inverse Laplace contour could not be lowered further due to merging poles with $\Im(\omega_s) > 0$. We may now consider the case when we are able to put the inverse Laplace contour in the lower half ω plane as shown in figure 7.6 without two merging poles.

Since the all poles are in the lower ω half plane, we find from residue calculation that

$$\lim_{t\to\infty} P(x,t,y) = \frac{1}{2\pi} \lim_{t\to\infty} \int_{L'} \hat{P}(x,\omega,y) e^{-i\omega t} \mathrm{d}\omega = 0, \tag{7.36}$$

meaning that the response decays for long time at every <u>finite</u> point x. However, as can be seen in figure 7.6, there may be an $\alpha_u \in \mathbb{C}$ which has crossed the real axis, thus for the sketched example hinting at spatial growth on the right hand side of the source for $x > 0$. The key point here is that even though at every fixed finite point (7.36) holds, it may well be that the response grows in x while it also travels along some characteristic or trajectory in the x direction - meaning that once it has passed a point x, the response at this point decays as $t \to \infty$. Then, even if (7.36) holds, we observe growth when following the response through the system.

For that purpose, Briggs considered the response in a frame moving with constant velocity by means of the Galilean transform

$$x = Vt, \tag{7.37}$$

bringing the response (7.27) to

$$P(V,t,y) = \frac{1}{4\pi^2} \int_{L'} \int_{F'} \hat{G}(\alpha,\omega,y) e^{i(\alpha V - \omega)t} \mathrm{d}\alpha \mathrm{d}\omega. \tag{7.38}$$

Two manipulations are now carried out. First, the inverse Fourier integral is calculated using the residue theorem as in (7.24), with the contribution of the pole α_u for $x > 0$ (see figure 7.6). Secondly, we are allowed to put the contour L' on the real ω axis without changing the outer integral, since no additional poles are enclosed by this procedure. We thus set ω equal its real part ω_r. The situation is sketched in figure 7.7. (7.38) thus becomes

$$P(V,t,y) = \frac{1}{2\pi} \int_{L'} \frac{i\, e^{i(\alpha_u(\omega_r)V - \omega_r)t}}{\left(\frac{\partial}{\partial \alpha} \left(\hat{G}(\alpha,\omega_r,y) \right)^{-1} \right) \Big|_{\alpha_u(\omega_r)}} \mathrm{d}\omega_r. \tag{7.39}$$

Note that the dependence on $\alpha_u(\omega_r)$ was explicitly kept for clarity. We focus on the exemplary consideration of the pole α_u here, as this pole leads to spatial growth away from the source, having crossed

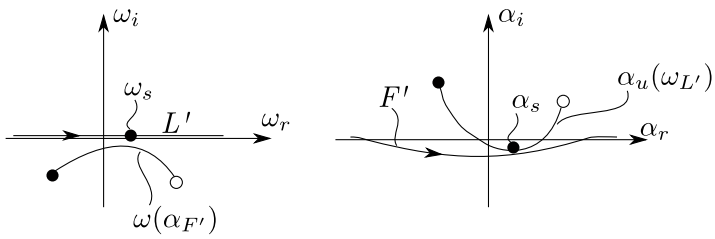

Figure 7.7.: Deformed inverse Fourier and Laplace contour in case of a convective instability in a moving Galilean frame. The saddle point is given at α_s.

the real axis. Therefore, the former equation holds for $x > 0$. If other poles cross the real α axis, their contribution has to be taken into account as well. The aim is now to evaluate the asymptotic behavior of (7.39) for $t \to \infty$, which is done by applying the saddle point method (see appendix A.6.5). We are therefore searching for so called saddle points of $i(\alpha_u(\omega_r)V - \omega_r)$ over ω_r, which are given by

$$0 = \frac{\mathrm{d}}{\mathrm{d}\omega_r}\left(\alpha_u(\omega_r)V - \omega_r\right) = \frac{\mathrm{d}\alpha_{u,r}}{\mathrm{d}\omega_r}V + i\frac{\mathrm{d}\alpha_{u,i}}{\mathrm{d}\omega_r}V - 1. \tag{7.40}$$

This condition requires

$$\frac{\mathrm{d}\alpha_{u,i}}{\mathrm{d}\omega_r} = 0, \tag{7.41}$$

which is, as can be seen in figure 7.7, fulfilled at $\omega = \omega(\alpha_s) = \omega_s$. (7.41) further yields for the observer velocity

$$V = \left(\frac{\mathrm{d}\alpha_{u,r}}{\mathrm{d}\omega_r}\right)^{-1}\Bigg|_{\omega=\omega_s}. \tag{7.42}$$

Note that the indices r and i again refer to the real and imaginary parts. From the saddle point method, we had to deform the path L' and in a way that it passes near ω_s. Then, considering the limit $t \to \infty$ of (7.39), as explained in appendix A.6.5, we use Laplace's method to find that for the response

$$\lim_{t\to\infty} P \sim t^{-\frac{1}{2}} e^{-\alpha_{u,i}(\omega_s)Vt} \tag{7.43}$$

holds. Here, $\alpha_{u,i}(\omega_s) < 0$ and is, we can see in figure 7.7, the maximum spatial growth rate for $\omega \in \mathbb{R}$ right of the source.

We conclude that even if the response decays at fixed positions x, if we move with observer velocity (7.42), we observe an unstable perturbation traveling through the system. This is exactly the nature of a convective instability. Summarizing, such instabilities are found as follows: Lowering the inverse Laplace contour L to the real axis with $\omega \to \omega_r$, the root $\alpha_u(\omega_r)$ describes a convective instability for $x > 0$, if it has crossed the real α axis and now has imaginary part $\alpha_{u,i} < 0$. For $x < 0$, conversely, we require a root $\alpha_l(\omega_r)$ to cross the real α axis from the lower half plane and thus have $\alpha_{l,i} > 0$. The saddle point describing the maximum growth of this convective instability is described by ω_s and can be determined by equation (7.41). From the dispersion relation for real frequencies ω_r, we further obtain the propagation velocity V using equation (7.42).

7.2.4. Harmonic source with finite extend leading to convective instabilities

The considerations carried out up to this point were quite general and mainly analyticity of the source function on the semicircle that was used to close the contour of integration was assumed, with the

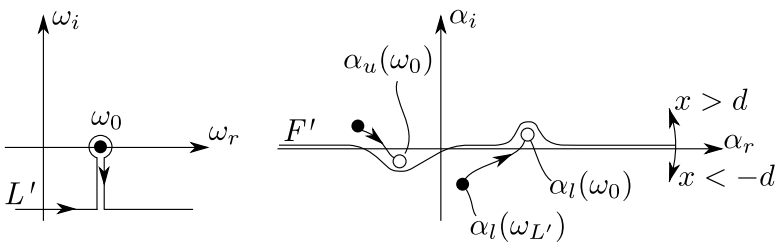

Figure 7.8.: Deformed inverse Laplace and Fourier contours for a harmonic source without absolute instabilities.

central outcome of two criteria determining convective and absolute instability. We may further ask, analogous to classical formulations of mass-spring-damper oscillations, whether a harmonic excitation might lead to a different stability behavior. Therefore, this section is intended to shed light on the excitation of an instability through a harmonic source in the absence of absolute instabilities.

A temporally harmonic source with frequency ω_0 is given in physical and Laplace space by

$$s_t(t) = e^{i\omega_0 t}, \qquad s_t(\omega) = \frac{1}{i(\omega - \omega_0)}. \tag{7.44}$$

Since the source should have finite extend in space, Briggs suggested to assume a source that is nonzero for $-d < x < d$, expressed by superposition of two Heaviside functions H. In physical and Laplace space this is given by

$$s_x(x) = \frac{1}{2d}\left(H(x+d) - H(x-d)\right), \qquad s_x(\alpha) = \frac{\sin(\alpha)}{\alpha d}. \tag{7.45}$$

Let us assume that we do not have absolute instabilities, meaning that we may lower the path of the inverse Laplace transform (7.18) to the real axis without violating causality as it was described in the previous section for the general case of a convective instability. However, due to the source (7.44), a fixed, non movable, pole on the real ω axis at $\omega_0 \in \mathbb{R}$ arises. To obtain a convective instability, again, some poles in the α plane must have crossed the real axis, as sketched in figure 7.8. There, the inverse Laplace contour is lowered into the lower complex ω half plane, however, in order to satisfy causality, the pole at $\omega = \omega_0$ must be enclosed. When varying ω from the lower half plane up to the source frequency ω_0, let us assume that the poles in the α plane move as shown. The inverse Fourier contour has to be deformed around these poles in order to keep the integrand analytic along F' (in other words to keep path homotopy without crossing any poles). Same as for the previous considerations of the inverse Fourier integral (7.21), the contour is closed for points outside the source, i.e. $|x| > d$. In order to apply Jordan's lemma, it is closed in the lower half plane for $x < -d$ and in the upper half plane for $x > d$ and poles which are enclosed contribute to the spatial behavior of the source response. Since both α_l and α_u have crossed the real α axis, their contribution due to the residue theorem leads to spatial growth for $x < -d$ due to α_l or for $x > d$ due to α_u.

The temporal behavior of the source response, included in the inverse Laplace transform (7.18) with deformed curve L', is for $t > 0$ determined by the pole at ω_0, thus leading to purely harmonic oscillating behavior. We can thus expect the source response being spatially amplified and harmonic in time. This is underlined by calculating the leading behavior of the source response in the following. We start with the inverse Laplace transform (7.18) along the contour L', closed for $t > 0$ in the lower complex ω half plane. Applying Jordan's lemma, the contribution of the pole at ω_0 is given by residue calculation as

$$P(x,t,y) \approx 2\pi i \mathrm{Res}_{\omega_0}\left(\frac{\hat{P}(x,\omega,y)s_t(\omega)e^{-i\omega t}}{2\pi}\right) = \hat{P}(x,\omega_0,y)e^{i\omega_0 t}, \tag{7.46}$$

where we used the source function (7.44) to obtain that

$$\mathrm{Res}_{\omega_0}\left(s_t(\omega)\right) = \mathrm{Res}_{\omega_0}\left(-i(\omega-\omega_0)^{-1}\right) = -i, \tag{7.47}$$

(Meyberg & Vachenauer, 2006). Since we have $\omega_0 \in \mathbb{R}$, it is clear that the source response (7.46) is oscillatory in time. To assess its spatial behavior, which is included in $\hat{F}$ and has already been discussed by means of figure 7.8, we analyze the inverse Fourier integral (7.21) for the spatial source (7.45). It is given by

$$\hat{P}(x,\omega,y) = \int_{F'} \hat{G}(\alpha,\omega,y)\frac{\sin(\alpha)}{\alpha d}e^{i\alpha x}\mathrm{d}\alpha. \tag{7.48}$$

As discussed, closures of the contour for $|x| > d$ are shown in figure 7.8 . Consequently, the previous integral is determined by the residues of the enclosed poles of $\hat{G}$. For the situation sketched, these are $\alpha_l(\omega_0)$ in case $x < -d$ and $\alpha_u(\omega_0)$ for $x > d$. Analogous to (7.24), we obtain

$$\begin{aligned}\hat{P}(x,\omega,y) = &\frac{i\,e^{i\alpha_u(\omega_0)x}}{\left(\frac{\partial}{\partial\alpha}\left(\hat{G}(\alpha,\omega,y)\right)^{-1}\right)\Big|_{\alpha_u(\omega_0)}}\cdot\frac{\sin(\alpha_u(\omega_0))}{\alpha_u(\omega_0)d}\cdot H(x-d)\\ &-\frac{i\,e^{i\alpha_l(\omega_0)x}}{\left(\frac{\partial}{\partial\alpha}\left(\hat{G}(\alpha,\omega,y)\right)^{-1}\right)\Big|_{\alpha_l(\omega_0)}}\cdot\frac{\sin(\alpha_l(\omega_0))}{\alpha_l(\omega_0)d}\cdot H(d-x),\end{aligned} \tag{7.49}$$

where the Heaviside function H activates the solution branches for $x > d$ or $x < -d$. Due to the poles having crossed the real α axis once ω is varied towards $\omega_0 \in \mathbb{R}$, the response grows away from the source. It is therefore a convective instability with oscillatory character in time and exponential growth in space away from the source.

Summarizing, the process of finding such harmonically excited convective instabilities in the absence of absolute ones can, according to Briggs, be carried out as shown in figure 7.8: We start with finding the branches $\alpha(\omega_L')$ and classify them as α_u or α_l depending on the sign of their imaginary part. In a second step, we deform the inverse Laplace contour to the real axis, i.e. we vary $\omega_{L'} \to \omega_r$. Due to this variation of ω, those poles (i.e. solutions of the dispersion relation) that cross the real α axis from below are spatially amplifying waves for $x < -d$ and those crossing from above are spatially amplifying for $x > d$.

7.3. An outlook regarding the application to the hyperbolic tangent shear layer

The former considerations were quite general, without assuming a specific type of flow. The key advantage of Briggs' method is the idea that without explicitly constructing a Green's function, statements on the asymptotic long term behavior of a source response can be made just by considering the eigenvalues of the homogeneous problem. We will in the current section introduce the first step of such an analysis for the hyperbolic tangent shear layer considered in the present thesis. Since this is a subject of ongoing research and requires extensive calculations, we present concrete results only for the first step of a spatial stability analysis at $M = 3$ and give an outlook on what follows.

The first step is to place the inverse Fourier and Laplace contours F and L analogous to what was schematically shown in figure 7.2a. The inverse Laplace contour L is placed above all eigenvalues of the dispersion relation emerging from the inverse Fourier contour F, given on the real α axis. As explained

Since for USO solutions it holds that $2c_x - 1 < 0$, the leading order terms suggest a reflection coefficient $|R| < 1$ if $\omega > 0$ and $|R| > 1$ if $\omega < 0$. This is consistent with the observations made for this phase velocity range in figs. 5.14a, 5.15a and 5.16a.

In conclusion, we want to state at this point that the leading order analysis does not necessarily capture the full dynamics at the critical layer, especially for outstanding phenomena such as resonant over reflection and absorption of modes. Nonetheless, the sign change of (A.21) at $c_x = 0.5$ supports that wave scattering phenomena differ depending on both the phase velocity of the perturbation as well as the sign of the temporal frequency. As described, the behavior is different for solutions oscillatory in either the lower or the upper free stream.

A.5. Compatibility condition in a plane jet for 3D and 2D eigenvalues

The 3D equivalent of (2.18) is given by (3.45) including β as the spanwise wavenumber. When inserted into (3.45), the transformations (3.47) convert (3.45) to its 2D equivalent (2.18) and therefore form a set of equivalence transformations with respect to the differential equation.

The goal of the following consideration is to show that the compatibility conditions in the core of the jet are also invariant under the set of equivalence transformations. A central feature of the equivalence transformations is that the eigenfunction $\hat{p}$ is invariant. This is since invariance of the determining ODE is required. We consider the compatibility condition for PSR modes, for the 3D case following from the solution of (3.45) in the core as

$$\left.\frac{B_1}{B_2}\right|_{3\mathrm{D}} = e^{\sqrt{\alpha_{3\mathrm{D}}^2+\beta^2-M_{3\mathrm{D}}^2(\omega_{3\mathrm{D}}-\alpha_{3\mathrm{D}})^2}\Delta l_{3\mathrm{D}}}. \tag{A.32}$$

Inserting the equivalence transformations (3.47) brings it to

$$\left.\frac{B_1}{B_2}\right|_{2\mathrm{D}} = e^{\sqrt{\alpha_{2\mathrm{D}}^2-M_{2\mathrm{D}}^2(\omega_{2\mathrm{D}}-\alpha_{2\mathrm{D}})^2}\Delta l_{3\mathrm{D}}}. \tag{A.33}$$

Note that due to the discussed form invariance of the eigenfunctions, the amplitude ratio is also invariant under the transformations (3.47), meaning that when applying (3.47) we used that $\left.\frac{B_1}{B_2}\right|_{3\mathrm{D}} = \left.\frac{B_1}{B_2}\right|_{2\mathrm{D}}$. Comparing (A.33) with the 2D compatibility condition (6.7), we consequently find that for equivalence

$$\Delta l_{3\mathrm{D}} = \Delta l, \tag{A.34}$$

holds (where Δl denotes the inflection point distance for a 2D evolving mode).

We thus have shown that 3D evolving eigenvalues are to be obtained from 2D ones at lower Mach numbers but at the same inflection point distance. A completely equivalent consideration can be carried out for the LTW with the corresponding compatibility condition (6.7), leading to the same result.

A.6. Tools from Fourier/Laplace transformations and complex analysis

A.6.1. Convolution theorem

We consider the convolution integral, defined by

$$F(x) = (f * g)(x) = \int_{-\infty}^{\infty} g(x-\tau)f(\tau)\mathrm{d}\tau. \tag{A.35}$$

If we apply Fourier transform (the same theorem holds for Laplace transform) on both sides as defined by (7.12), we obtain

$$F(\alpha) = g(\alpha)f(\alpha), \tag{A.36}$$

i.e. the convolution is transferred to a multiplication in Fourier space (Meyberg & Vachenauer, 2006). This can be seen if formally Fourier transform is applied on (A.35), giving

$$\begin{aligned} F(\alpha) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x-\tau)f(\tau)e^{i\alpha x}\mathrm{d}\tau\mathrm{d}x \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x-\tau)f(\tau)e^{i\alpha(x-\tau)}e^{i\alpha\tau}\mathrm{d}\tau\mathrm{d}x, \end{aligned} \tag{A.37}$$

allowing us to seperate the integral. This is done by introducing a new set of variables, given by $\tilde{x} = x - \tau$ and $\tilde{\tau} = \tau$, thus leading to $\mathrm{d}\tau\mathrm{d}x = \mathrm{d}\tilde{\tau}\mathrm{d}\tilde{x}$. This brings the Fourier-transformed convolution integral into a separable form,

$$F(\alpha) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(\tilde{x})f(\tilde{\tau})e^{i\alpha\tilde{x}}e^{i\alpha\tilde{\tau}}\mathrm{d}\tilde{\tau}\mathrm{d}\tilde{x} = \int_{-\infty}^{\infty} g(\tilde{x})e^{i\alpha\tilde{x}}\mathrm{d}\tilde{x}\int_{-\infty}^{\infty} f(\tilde{\tau})e^{i\alpha\tilde{\tau}}\mathrm{d}\tilde{\tau}, \tag{A.38}$$

which obviously leads, using the definition of the Fourier transform (7.12), to (A.36). The same theorem can be shown to hold for the Laplace transform (7.13).

A.6.2. Residual theorem

We consider a simply connected domain in the complex plane, given by $D \subseteq \mathbb{C}$. Further, f is a function holomorphic on D_0. D_0 is given by D except for non-essential singularities $a_1, \ldots, a_n$ of f, i.e. $D_0 = D \setminus \{a_1, \ldots a_n\}$. Letting Γ be a rectifiable curve in in D_0 enclosing the singularities. Then, the line integral of f around Γ is given by (Meyberg & Vachenauer, 2006)

$$\oint_{\Gamma} f(z)dz = 2\pi i \sum_{k=1}^{n} \mathrm{ind}_{\Gamma}(a_k)\mathrm{Res}_{a_k}\left(f(z)\right). \tag{A.39}$$

Here, $\mathrm{Res}_{a_k}\left(f(z)\right)$ is the residue of f at the singularity a_k and ind_{Γ} is the winding number of Γ around a_k. For a positive oriented simple closed curve it holds that $\mathrm{ind}_{\Gamma} = 1$ whereas $\mathrm{ind}_{\Gamma} = -1$ for a negative oriented curve.

Importantly, if no singularities are enclosed or if the singularities are removable, the value of the line integral (A.39) vanishes. A special case for the residual theorem occurs if the function has a simple pole, i.e. if $f(z) = \frac{g(z)}{z-a}$ with g being holomorphic on D. We then obtain Cauchy's integral formula

$$\oint_{\Gamma} \frac{g(z)}{z-a}dz = 2\pi i\ \mathrm{ind}_{\Gamma}(a)g(a). \tag{A.40}$$

A.6.3. Jordan's lemma

Together with the residual theorem, Jordan's lemma is often used in order to calculate line integrals (Meyberg & Vachenauer, 2006). It makes a statement on line integrals in the complex half plane which structurally occur when closing contours in Fourier or Laplace transforms. We consider for $\lambda \geq 0$ the integrals in the complex plane

$$I_+ = \int_{\Gamma_{J+}} h(z)e^{i\lambda z}dz, \qquad I_- = \int_{\Gamma_{J-}} h(z)e^{-i\lambda z}dz, \tag{A.41}$$

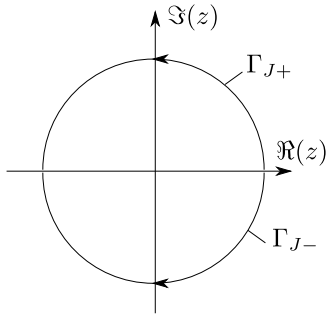

Figure A.1.: Open semicircles in the complex half plane referring to Jordan's lemma.

with the curves Γ_{J+} and Γ_{J-} having $|z| \to \infty$ as shown in figure A.1. If $h(z) \to 0$ in the upper and lower half plane on $|z| \to \infty$ respectively (or if $\lambda = 0$ and $z\,h(z) \to 0$), we have

$$I_+ \to 0, \qquad I_- \to 0. \tag{A.42}$$

Note that for $\lambda < 0$, the curves Γ_{J+} and Γ_{J-} in (A.41) have to be exchanged with each other.

A.6.4. Green's function for ODEs of second order

In this appendix we discuss in more detail the connection between the poles of Green's function and the dispersion relation. The discussion made in this appendix is oriented on the lecture notes of Skinner (2015). We restrict ourselves to the case of a second-order linear differential operator here, since this represents the problem treated by the PBE (2.18) in this work. The operator is denoted by $\mathcal{L}$ acting on $[a, b]$ (including also the case of an infinite interval $(-\infty, \infty)$ as for the shear layer). $\mathcal{L}$ defines the differential equation

$$\mathcal{L}p(y) = f(y), \tag{A.43}$$

where the forcing term f should be bounded on $[a, b]$. $\mathcal{L}$ may be written as

$$\mathcal{L} = a(y)\frac{\mathrm{d}^2}{\mathrm{d}y^2} + b(y)\frac{\mathrm{d}}{\mathrm{d}y} + c(y). \tag{A.44}$$

Green's function G referring to the operator $\mathcal{L}$ is defined as the solution to the problem

$$\mathcal{L}G(y, y_0) = \delta(y - y_0), \tag{A.45}$$

which satisfies homogeneous boundary conditions $G(a, y_0) = G(b, y_0) = 0$. With knowledge of a Green's function, we have the solution of (A.43) given by

$$p(y) = \int_a^b G(y, y_0) f(y_0) \mathrm{d}y_0. \tag{A.46}$$

The previous result is easily proven by applying $\mathcal{L}$ on (A.46), leading to

$$\mathcal{L}p(y) = \int_a^b \left(\mathcal{L}G(y, y_0)\right) f(y_0)\mathrm{d}y_0 = \int_a^b \delta(y - y_0) f(y_0)\mathrm{d}y_0 = f(y), \tag{A.47}$$

and thus yielding (A.43) again. Note that we used (A.45) and the sifting property of the delta distribution here. We note as a central result that a Green's function is independent of the forcing f but depends only on the linear operator and domain.

Next, we construct Green's function, revealing why solutions of the dispersion relation lead to poles of Green's function. From (A.45), we find that for all points $y \neq y_0$, Green's function is nothing but the solution of the homogeneous problem. For this homogeneous problem $\mathcal{L}p(y) = 0$, we may always find a basis of solutions $\{p_1(y), p_2(y)\}$ that with each of the basis solutions fulfilling one homogeneous boundary conditions by

$$p_1(a) = 0, \qquad p_2(b) = 0. \tag{A.48}$$

Thus, outside of y_0, we have

$$G(y, y_0) = \begin{cases} A(y_0)\, p_1(y), & \text{for} \quad y \in [a, y_0), \\ B(y_0)\, p_2(y), & \text{for} \quad y \in (y_0, b], \end{cases} \tag{A.49}$$

with $A(y_0)$, $B(y_0)$ being constant with respect to y and have to be determined. The determination is based on the requirement that G should be continuous at y_0. For details, the reader is referred to the derivation in Skinner (2015), including the result that a jump of $\partial G/\partial y$ at $y \to y_0$ occurs. Having determined the constant, we go back into (A.49) and obtain

$$G(y, y_0) = \begin{cases} \frac{p_2(y_0)}{a(y_0)W(y_0))}\, p_1(y), & \text{for} \quad y \in [a, y_0), \\ \frac{p_1(y_0)}{a(y_0)W(y_0))}\, p_2(y), & \text{for} \quad y \in (y_0, b]. \end{cases} \tag{A.50}$$

Here, a refers to the coefficient of the highest order derivative in $\mathcal{L}$ as defined by (A.44). Most important, W is the Wronskian defined by

$$W(y) = p_1(y)\, p_2'(y) - p_2(y)\, p_1'(y), \tag{A.51}$$

which is in (A.50) evaluated at y_0. The crucial point is now that W vanishes exactly at the eigenvalues of $\mathcal{L}$ (Teschl, 2012). The reason for this is that for an eigenvalue, each solution branch p_1 and p_2 fulfills both homogeneous BCs, thus they become linearly dependent, i.e. for an eigenvalue of $\mathcal{L}$ it holds that $p_2(y) = C\, p_1(y)$. Inserting this into (A.51) consequently leads to $W = 0$ for all eigenvalues.

With that result, it becomes clear why eigenvalues of the problem, which are solutions of the dispersion relation at the same time, lead to singularities of Green's function that have to be taken in consideration when integrating in both the complex α and the complex ω - plane as shown in chapter 7.

A.6.5. Method of stationary phase/steepest descent

The method of stationary phase is used to approximate a contour integral in the complex plane (Chaichian & Demichev, 2001). In the context of chapter 7, it was used to determine the long-term behavior of an integral which represents the answer to an external source. The integral generally has the form

$$I = \int_C h(z) e^{\lambda\, g(z)} \mathrm{d}z, \tag{A.52}$$

and we are interested in its behavior for $\lambda \to \infty$. We assume g and h to be holomorphic. Then, we are allowed to deform the curve C without changing the value of the integral to a new curve C' which

fulfills (i) $\Im(g(z)) = const. = g_i$ and (ii) passes through points a global maximum of g at z_0 given by $\mathrm{d}g/\mathrm{d}z = 0$. From condition (i), we may rewrite I as

$$I = e^{i\lambda g_i} \int_{C'} h(z) e^{\lambda \Re(g(z))} \mathrm{d}z, \tag{A.53}$$

which we may now apply Laplace's method (Erdélyi, 1956) to. Due to condition (ii), we expand $\Re(g(z)) = g_r(z)$ as Taylor series around the maximum z_0, yielding

$$g_r(z) \approx g_r(z_0) - \frac{1}{2} |g_r''(z_0)|(z - z_0)^2, \tag{A.54}$$

where it is already incorporated that the second derivative is negative due to the maximum property. With that, (A.53) becomes

$$I \approx e^{i\lambda g_i} e^{\lambda g_r(z_0)} \int_{C'} h(z) e^{-\frac{\lambda}{2} |g_r''(z_0)|(z-z_0)^2} \mathrm{d}z. \tag{A.55}$$

The remaining integral is of Gaussian type, which may be, according to Erdélyi (1956), for large values of λ approximated to finally yield

$$I \approx e^{i\lambda g_i} e^{\lambda g_r(z_0)} h(z_0) \sqrt{\frac{2\pi}{\lambda |g_r''(z_0)|}}\,. \tag{A.56}$$

Since g_i is constant, we summarize to

$$I \approx e^{\lambda g(z_0)} h(z_0) \sqrt{\frac{2\pi}{\lambda |g_r''(z_0)|}}\,, \tag{A.57}$$

where we have to recall that z_0 lies on the deformed curve C' where the phase of g is stationary (i.e. the imaginary part is constant).